COMPARATIVE ANALYSIS SARS COV-2 SEQUENCES ISOLATED IN FRANCE AND NIGERIA

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ABSTRACT

Reports have it that in the last corona virus SARS-Cov-2 pandemic have shown more severity in the Americas and European countries as compared to sub-saharan Africa. Various reasons have been reported to have been responsible. However, monitoring the last pandemic regards to geographical and evolutionary distribution is important to understand better the pathogenicity and transmissibility through its featured genome and proteins responsible for its virulence geographically. All the genome sequences isolated in Nigeria and equal number of genome sequences from France obtained from BV-BRC resource were comparatively analysed. Phylogenetic tree analysis was carried out using the Virus Tree at BV-BRC.Multiple sequence alignment of these sequences were carried out on all the genomes and compared with each other using the MAS programs available at BV-BRC resource. Genome analysis and variability with featured protein was also analysed. From the phylogenetic tree analysis it showed that the Nigeria tree has several clusters showing that SARS CoV-2 may have entered Nigeria from different countries as compared to Sequences isolated from France. However, the Nigerian Sequences showed a marked clusters relative to Wuhan sequence. A long the line most sequences actually diverged form the Wuhan sequence because of the genetic distance relatedness to those from France. Protein analysis revealed that some ORF 3b, 3d and 9c were not expressed which could be reasons for low virulence in sub-saharan Africa as the virus crosses the geographical barrier lost isits power to express those essential proteins responsible for its virulence.It is important to note that there was a difference in the structural proteins of SARS CoV-2 from both regions in the spikes suggesting a divergence in the tropical Africa which reduces the motif binding ability to receptor sites as a result of deletions as the virus crosses geographical barriers. This was noticed in the ORF 3b, 3d and 9c in the spike protein. This explains the reduced virulence in sub-Saharan Africa. And severity of the pandemic in Europe. There was a cluster of dissimilar biding motifs in France which were opposite of that diverged in Nigeria. This could be a source of drug target for Covid-19.

INTRODUCTION

The last SARS-CoV-2 pandemic which started in Wuhan China is the 7th human corona virus infection with devastating proportion with over 437 million people affected and 5.9 million deaths. The outbreak which started in Wuhan China was reported as an outbreak of pneumonia by the municipal health commission of Wuhan by 31st December 2019 (1). The disease which was named corona virus disease 2019 (COVID-19) by WHO of which International Committee on Taxonomy recognized it as a new infectious agent SARS CoV-2 (2). The virus spread globally with devastating effects with various variants of concern emerging with America, Asia and Europe worse hit with Africa accounting for the lowest number of cases and deaths (3,4,5) The severity, virulence and transmissibility of the outbreak in Europe far out ways both morbidity and mortality that was obtained in Africa. It has now become imperative to determine what was responsible for low severity of the outbreak in Africa which could be a target for drug discovery.

SARS CoV-2 which is a member of the genus beta corona virus (beta-CoV) with positively sensed RNA genome of 30kb with define 13 open reading frames (ORF) coding for structural, no-structural and accessory proteins (6,7). The four structural proteins are spike (S), envelope (E), membrane (M) and nucleocapsid (N). The S protein consist of 2 structural subunit S1with angiotensin converting enzyme 2 (ACE 2) receptor binding domain (RBD) as well as S2 subunits which possess element for membrane fusion. The M protein is a trans-membrane glycoprotein crucial for membrane fusion, whereas the E protein is required for virion assembly and morphogenesis (7). In case of SARS-CoV-1, the N protein which is highly antigenic is used as a serological marker (8). The nonstructural replicase proteins (pp1a: nsp1–nsp11 and pp1b: nsp12–nsp16) are involved in mRNA synthesis and replication, and the accessory proteins (3a, 3b, 6, 7a, 7b, 8 and 9b) participate in modulating host innate immunity. Upon infection, SARS-CoV-2 first gets attached to the naso-/oro-pharyngeal inner linings and then moves down to the lungs, which are even richer in ACE2, and triggers cell damage.

MATERIALS AND METHODS

All the sequences used for this analysis were downloaded from the Bacterial Viruses Bioinformatics Resource (BV-BRC) where all the analysis was also done. A structured literature search was first carried out with keywords corona virus, SARS-CoV-2 transmissibility and pathogenicity, SARS-Cov-2 epidemiology, SARS CoV-2 genome comparative analysis. All 245 SARS-CoV-2 genomic sequences from Nigeria obtained in year 2020 were downloaded as well as 400 SARS CoV-2 genomic sequences isolated from France. Nigeria was used as a representative for sub-saharan Africa while France was used to represent Europe. A Wuhan genomic sequence isolated in 2019 was also downloaded and used as reference genomic sequence. All sequences were downloaded and left in the workspace at BV-BRC server before the analysis.

Phylogenomics analysis

In other to determine the ancestral relationships of all the sequences phylogenetic analysis of all the sequences using the Virus Tree tool in BV-BRC server were carried out on sequences from France using the Wuhan sequence as reference sequence. Same was done for the Nigerian sequences. Tree Analysis of all the sequences the sequences combine was also carried out. The alignment was done with mafft on the BV-BRC server.

Genome analysis

All the genome analysis which included the assembly and annotation was also carried out using the SARS CoV-2 analysis tool in BV-BRC. Data analysis and output was done in Pangolin.

Protein analysis

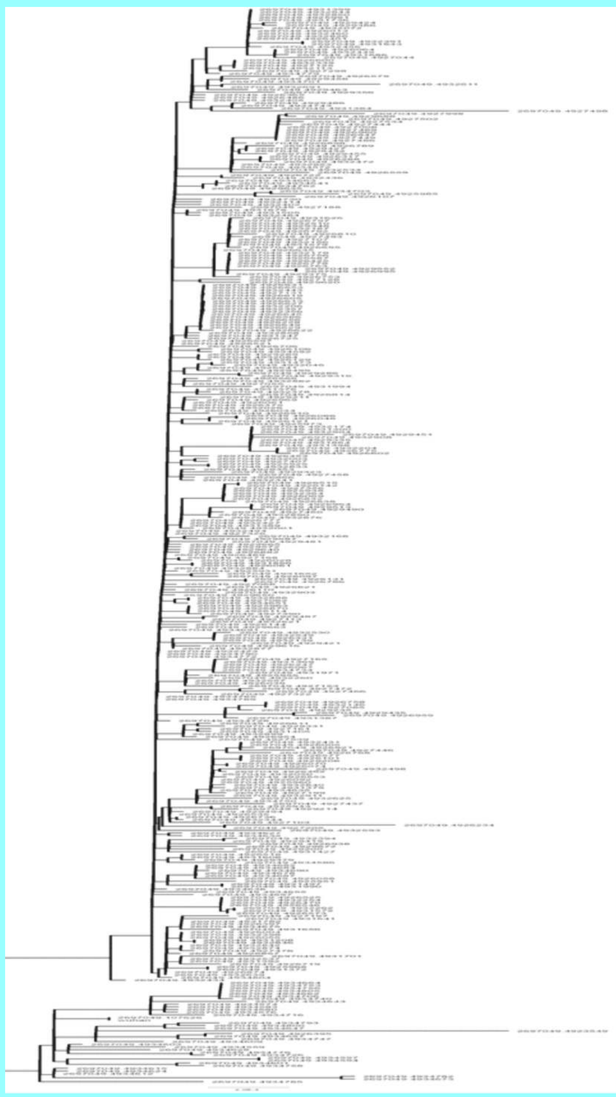
As a featured group on the BV-BRC all corresponding protein as well as the ORFs were analyze and the differences in ORFs and proteins were determined.

Genome Alignment

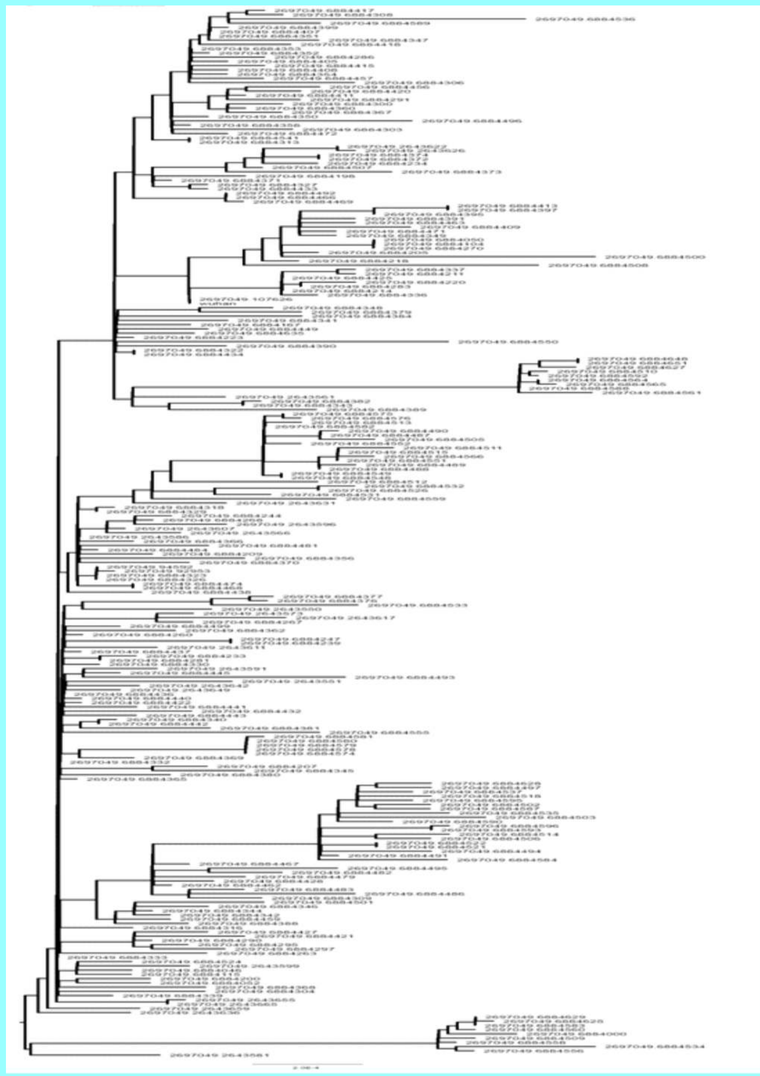
Genome alignment was carried out using the Mauve a genome alignment tool available at BV-BRC and all the genomes were aligned to determine mutational domains and related domains that have been transferred. Alignments were analysed using genedoc.

RESULTS

Phylogenetic tree analysis revealed that Most SARS CoV-2 genomic sequences originated from the Wuhan Human SARS CoV-2 sequence isolated in 2019. 63 of the sequences had 100% identity with the Wuhan sequence and may be refereed as homologous (Fig. 1).The three major clades on the tree all originated from the Wuhan sequence withshorter evolutionary distance with very few evolutionary grouping and divergence. Unlike the Nigerian tree which has longer evolutionary distance from the Wuhan sequence.Though there are possibilities of same ancestral origin, but the Nigerian sequences have diverged over time and looked closer to the sequences from France unlike that from Wuhan. A clade that diverged directly from the Wuhan sequence did produce any further evidence of divergence beyond that clade in the Nigerian tree (Fig. 2).There were several tree trunks in the Nigerian tree which gave the likelihood that most SARS CoV-2 sequences in Nigeria was not possibly from a particular region.



Genomic tree analysis of the genomic sequences from France. The tree shows that the mothers ancestor of all SARS CoV-2 sequences were rooted from the Wuhan sequence isolated in 2019. 63 of the sequences were 100% identical and on the tree they were in the same clade.



Virus tree analysis of 245 SARS CoV-2 sequences isolated from Nigeria. There were marked differences in their ancestral relationship between the sequences in reference to the reference Wuhan sequence. The tree shows a high degree of divergence from the Wuhan reference sequence as though a few of the Nigerian sequences were in the same clade with the Wuhan reference sequence unlike that of France where several or more than ¾ of all the France sequences had the same ancestral origin as that of the Wuhan.

REGION	POS	REF	ALT	REF_DP	REF_RV	REF_QUAL	ALT_DP	ALT_RV	ALT_QUAL	ALT_FREQ	TOTAL_DP	PVAL	PASS	GFF_FEATURE	REF_CODON	REF_AA	ALT_CODON	ALT_AA
MN908947.3	241	C	T	21	0	93	432	0	93	0.953642	453	1.35308e-235	TRUE	NA	NA	NA	NA	NA
MN908947.3	3037	C	T	20	0	93	449	0	93	0.957356	469	5.37099e-246	TRUE	cds-YP_009724389.1	TTC	F	TTT	F
MN908947.3	3037	C	T	20	0	93	449	0	93	0.957356	469	5.37099e-246	TRUE	cds-YP_009725295.1	TTC	F	TTT	F
MN908947.3	14408	C	T	21	0	93	442	0	93	0.954644	463	3.90597e-241	TRUE	cds-YP_009724389.1	CCT	P	CTT	L
MN908947.3	23403	A	G	21	0	93	439	0	93	0.954348	460	2.18002e-239	TRUE	cds-YP_009724390.1	GAT	D	GGT	G

Major source of variation and mutation were mainly noticed in the S proteins with the RBD showing most of the variability in sequences from both countries.

Genome Analysis

Mutational analysis of both between the Nigerian and France sequences reveal some areas of SNP along a domain. The S gene show a mutational difference at D614G, while it occurred at ORF 1ab at F924F, F924F and P314L the Pangolin version 4.3.1 shows that all the sequences belong to lineage B.1 with a conflict of 0.4782608695652174.

Variation/ mutational analysis

Major source of variation and mutation were mainly noticed in the S proteins with the RBD showing most of the variability in sequences from both countries.

ID	Start	Strand	Length	Function
fig|2697049.9243527.CDS.1	266	+	13218	ORF1a polyprotein
fig|2697049.9243527.CDS.2	266	+	13203	ORF1ab polyprotein
fig|2697049.9243527.mat_peptide.1	266	+	540	leader protein
fig|2697049.9243527.mat_peptide.2	806	+	1914	nsp2
fig|2697049.9243527.mat_peptide.3	2720	+	5835	nsp3
fig|2697049.9243527.mat_peptide.4	8555	+	1500	nsp4
fig|2697049.9243527.mat_peptide.5	10055	+	918	3C-like proteinase
fig|2697049.9243527.mat_peptide.6	10973	+	870	nsp6
fig|2697049.9243527.mat_peptide.7	11843	+	249	nsp7
fig|2697049.9243527.mat_peptide.8	12092	+	594	nsp8
fig|2697049.9243527.mat_peptide.9	12686	+	339	nsp9
fig|2697049.9243527.mat_peptide.10	13025	+	417	nsp10
fig|2697049.9243527.mat_peptide.11	13442	+	39	nsp11
fig|2697049.9243527.mat_peptide.12	13442	+	2796	RNA-dependent RNA polymerase
fig|2697049.9243527.mat_peptide.13	16238	+	1803	helicase
fig|2697049.9243527.mat_peptide.14	18041	+	1581	3'-to-5' exonuclease
fig|2697049.9243527.mat_peptide.15	19622	+	1038	endoRNAse

A feature of reports showing areas of variabilitywhen side by side. The Nigerian SARS CoV-2 sequences in Nigeria shows high degree of variability compared to the France sequences with most of the domains showing no evidence of expression.

DISCUSSION

The phylogenetic tree analysis showed that most sequences from France originated from the Wuhan sequence thereby concluding that most of the B.1 lineage seen in France in 2020 actually originated from China. In the Nigerian Sequences there were several clusters with only one cluster representing the Wuhan sequences which is a pointer to the fact that carriers actually entered Nigeria from different countries but the marked difference in both sequence and protein mutational analysis showed that sequences crossing the geographical barriers actually mutated and became less virulence and less transmissible. This was noticed when proteins featured in the France sequences showed no evidence of expression in the Nigeria sequences. These proteins were in the ORF 3d,3b, 9c and of which all were responsible for SARS CoV-2 virulence. Further analysis of the mutational changes in this open reading frames could be a target for drug discovery.

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